

Feature Engineering and Feature Selection Report

1. Project Overview

This project focuses on preparing the Titanic dataset for machine learning by performing feature engineering and feature selection. The goal is to transform raw data into a format suitable for predictive modeling and identify the most relevant features.

2. Objective

The objectives of this project are:

- Handle missing values in the dataset.
 - Convert categorical variables into numerical values.
 - Scale numerical features.
 - Select the most relevant features for machine learning.
 - Generate a processed dataset for future model training.
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3. Dataset Description

- **Dataset Name:** Titanic Passenger Dataset
 - **File Format:** CSV
 - **Number of Records:** 891
 - **Target Variable:** Survived
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4. Feature Engineering Process

a) Handling Missing Values

- Missing values in the **Age** column were replaced with the median value.
- Missing values in the **Embarked** column were replaced with the most frequent value (mode).
- The **Cabin** column was removed because it contained many missing values.

b) Encoding Categorical Features

Categorical columns cannot be directly used by machine learning algorithms. Therefore:

- **Sex** was converted into numerical values.
- **Embarked** was converted into numerical values using Label Encoding.

c) Feature Scaling

The numerical columns **Age** and **Fare** were standardized using **StandardScaler** to ensure that all numerical features are on a similar scale.

5. Feature Selection

The **SelectKBest** method with the **Chi-Square (Chi²)** statistical test was used to select the most relevant features.

Selected Features

- Sex
- Age
- SibSp
- Parch
- Embarked

These features showed the strongest relationship with the target variable (**Survived**).

6. Output

The processed dataset was saved as:

`engineered_features.csv`

This dataset contains only the selected features along with the target variable.

7. Benefits of Feature Engineering

- Improved data quality.
 - Converted categorical data into numerical format.
 - Standardized numerical features.
 - Reduced unnecessary information.
 - Prepared the dataset for machine learning models.
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8. Challenges Faced

- Understanding which features should be encoded and scaled.
 - Handling missing values without losing important information.
 - Selecting the most relevant features using statistical methods.
 - Organizing the processed dataset and project files for GitHub submission.
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9. Conclusion

Feature engineering and feature selection are essential preprocessing steps in machine learning. In this project, missing values were handled, categorical features were encoded, numerical features were scaled, and the most relevant features were selected. These steps help improve the quality of the dataset and prepare it for building efficient machine learning models.